

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch: Civil Engineering Semester :V Subject Code & Name: BTHM505 Project Management Max Marks: 60 Date: 14/02/2023 Duration: 3 Hr.																																									
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																																									
											(Level/CO)	Mark s																														
Q. 1	Solve Any Two of the following.											12																														
1.A)	Explain Bar chart, its aspects, advantages and dis-advantages.										CO1	6																														
1.B)	Draw the network and calculate TE & TL for all activates for the following <table border="1"><tr><td>Activity (i-j)</td><td>1-2</td><td>2-3</td><td>2-4</td><td>3-5</td><td>3-6</td><td>4-5</td><td>4-7</td><td>5-8</td><td>6-8</td><td>7-8</td></tr><tr><td>Duration (t^{ij}) Days</td><td>5</td><td>2</td><td>6</td><td>4</td><td>4</td><td>2</td><td>3</td><td>7</td><td>8</td><td>2</td></tr></table>										Activity (i-j)	1-2	2-3	2-4	3-5	3-6	4-5	4-7	5-8	6-8	7-8	Duration (t^{ij}) Days	5	2	6	4	4	2	3	7	8	2	CO1	6								
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Duration (t^{ij}) Days	5	2	6	4	4	2	3	7	8	2																																
1.C)	Prepare table and calculate EST, EFT,LST,LFT, and Total Float associated with the project in Q.no.1(B) , Find critical path & project Duration										CO1	6																														
Q.2	Solve Any Two of the following.											12																														
2.A)	Explain briefly Direct cost and indirect cost for construction project										CO2	6																														
2.B)	The following table gives the data for the duration and costs of each activity of project network. The indirect cost of the project 3000Rs/Week. <table border="1"><tr><td>Activity</td><td>Normal Duration (Week)</td><td>Normal Cost (Rs.)</td><td>Crash Duration (Week)</td><td>Normal Cost (Rs.)</td></tr><tr><td>1-2</td><td>6</td><td>7000</td><td>3</td><td>14500</td></tr><tr><td>1-3</td><td>8</td><td>4000</td><td>5</td><td>8500</td></tr><tr><td>2-3</td><td>4</td><td>6000</td><td>1</td><td>9000</td></tr><tr><td>2-4</td><td>5</td><td>8000</td><td>3</td><td>15000</td></tr><tr><td>3-4</td><td>5</td><td>5000</td><td>3</td><td>11000</td></tr></table> Draw network. Find project duration and show critical path based on normal duration and corresponding total project cost. Calculate cost slope for each activity.										Activity	Normal Duration (Week)	Normal Cost (Rs.)	Crash Duration (Week)	Normal Cost (Rs.)	1-2	6	7000	3	14500	1-3	8	4000	5	8500	2-3	4	6000	1	9000	2-4	5	8000	3	15000	3-4	5	5000	3	11000	CO2	6
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2.C)	Determine the optimum duration of project and the corresponding minimum cost associated with the project in Q.no.2 (B)	CO2	6																																
Q. 3	Solve Any Two of the following.		12																																
3.A)	Differentiate between CPM and PERT	CO3	6																																
3.B)	Draw the network and find the project duration & Critical path for the following	CO3	6																																
	<table> <tr> <th>Activity (i-j)</th><th>t_o (Optimistic Time)</th><th>t_L (Most likely Time)</th><th>t_p (Pessimistic Time)</th></tr> <tr> <td>1-2</td><td>6</td><td>9</td><td>18</td></tr> <tr> <td>1-3</td><td>5</td><td>8</td><td>17</td></tr> <tr> <td>2-4</td><td>4</td><td>7</td><td>22</td></tr> <tr> <td>3-4</td><td>4</td><td>7</td><td>16</td></tr> <tr> <td>4-5</td><td>4</td><td>10</td><td>22</td></tr> <tr> <td>2-5</td><td>4</td><td>7</td><td>10</td></tr> <tr> <td>3-5</td><td>2</td><td>5</td><td>8</td></tr> </table>	Activity (i-j)	t _o (Optimistic Time)	t _L (Most likely Time)	t _p (Pessimistic Time)	1-2	6	9	18	1-3	5	8	17	2-4	4	7	22	3-4	4	7	16	4-5	4	10	22	2-5	4	7	10	3-5	2	5	8		
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3.C)	Find Standard Deviation and Z value if the project mentioned in Q. no 3 (B) has the completed in 35 Days	CO3	6																																
Q.4	Solve Any Two of the following.		12																																
4.A)	Explain Demand and supply	CO4	6																																
4.B)	Explain different types of interests	CO4	6																																
4.C)	Explain the importance of economics in the construction industry (Civil Engineering field).	CO4	6																																
Q. 5	Solve Any Two of the following.		12																																
5.A)	Explain the uses, advantages, and limitations of Break-even analysis.	CO5	6																																
5.B)	Explain the importance of Total Quality Management (TQM).	CO5	6																																
5.C)	Explain the uses of computer software (Microsoft Project and Primavera) in project management.	CO5	6																																
	*** End ***																																		

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